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Editor-in-Chief

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Dear Editor-in-Chief,

We submit for your consideration the manuscript:

**CCD-MAPPO: A Coupling Coordination Degree Signal
for Cooperative Multi-Agent Supply Chain Resilience**

Motivation and scope. Global supply chains face an underappreciated structural challenge: firms differ not only in *how much* data they exploit, but in whether they invest broadly across many data sources (breadth) or deeply in a few (depth). This breadth–depth heterogeneity, captured by the Coupling Coordination Degree (CCD) from coupling-coordination theory, shapes disruption recovery speed in ways that aggregate digitalization indices obscure. No prior multi-agent reinforcement learning (MARL) framework embeds this structural dimension into the state space or reward signal. Our paper closes that gap.

Contributions. We propose CCD-MAPPO, a MAPPO algorithm augmented by the CCD signal, with three principled theoretical contributions:

1. **Representational efficiency** (Proposition 1): no symmetric linear feature $\alpha B + \beta D$ can distinguish the balance non-linearity that CCD encodes; CCD’s canonical C/T factorization is analytically irreducible.
2. **Provable safety** (Proposition 2): a residual-policy construction anchored on the (s, S) heuristic guarantees worst-case performance at the classical baseline and preserves B2B information privacy at execution time.
3. **Linear scaling** (Proposition 3): a graph neural network critic with $O(|\mathcal{E}|)$ parameter complexity—vs. $O(|\mathcal{N}|^2)$ for MLP critics—enables deployment on industrial-scale supply networks.

Against five baselines over four disruption scenarios (100 rollouts each), CCD-MAPPO achieves a 4.1% improvement over vanilla MAPPO, 12.4% over QMIX, and 23.7% over the (s, S) heuristic, all confirmed at $p < 0.05$ by Wilcoxon rank-sum tests.

Fit with *Annals of Operations Research*. The paper combines stochastic dynamic optimisation, supply chain management, and MARL—an intersection central to AOR’s interdisciplinary scope. The safety guarantee and closed-form scaling result provide the analytical rigour expected of the journal, while the four-scenario experiment grounds the contributions in operations practice.

Declarations. This manuscript is original, has not been previously published, and is not currently under review at any other journal. The author declares no competing interests. Large language model tools were used solely for language editing and L^AT_EX formatting; all scientific content and conclusions are the sole responsibility of the author.

We are confident this work makes a meaningful contribution to the literature on data-driven supply chain resilience and respectfully ask that it be considered for publication in *Annals of Operations Research*.

Yours sincerely,

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